



Zertifikat

Ethical Hacking and Penetration Testing at University Level

Certificate Number: 2025-03-003

Luca Claus has successfully completed the module **Ethical Hacking**. The module comprised 4 ECTS credits, corresponding to a workload of **120 hours**. The **final examination**, consisting of a **practical hacking challenge**, the preparation of a **penetration testing** report, and a **written exam**, was completed with the overall grade:

1,5 (Very good)

The module provides practical and theoretical knowledge in **Ethical Hacking** and **Penetration Testing** at university level and was conducted in the Summer Semester 2025 by **Florian Dalwigk**.

Luca Claus has acquired knowledge in the following areas:

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|----------------------|-------------------|------------------------|--------------------|
| + Kali Linux | + Legal | + MITRE ATT&CK | + CVE/CVSS |
| + Cyber Kill Chain | Foundations | + XSS | + SQLInjection |
| + Buffer Overflows | + OWASP Top 10 | + Enumeration | + Password hacking |
| + Social Engineering | + Scanning | + Privilege Escalation | + AI in Ethical |
| + OWASP Top 10 | + Malware | + Operational | Hacking |
| LLM | + Writing Pentest | Security | + BurpSuite |
| + Hashcat | Reports | + Nmap | + Shodan |
| + Hydra | + Metasploit | + ShellGPT | + Ollama |
| + sqlmap | | | |

Karlsruhe, 26.03.2025

Vice Dean
Prof. Dr.-Ing. Holger Vogelsang